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Using device: mps
Using single-amplitude feyn data from /var/folders/1s/69py7bdx0g7g0t4_8rjh6g9h0000gn/T/single_amplitude_eval_rphf76s7/first_amplitude.csv
Imported 1 token rows
Wrote raw expression data to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testing/outputs/gravity5unified12345_seed_beam20.csv
Wrote normalized token data to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testing/outputs/gravity5unified12345_seed_beam20_tok.csv
Input token limit: unlimited; longest input: 899 content / 901 with BOS/EOS
Numerical evaluator: gravity (processes={'3s2h': 1}; reference_modes=first,last,random; gauge_shift=True; tol_abs=2e-09; tol_rel=2e-08)
Loading model from /Users/kymani/Documents/Big_YM_Data_Sets/Cyclic_Gravity/Model/best_model.pt

Running decode mode: beam
  method      : beam
  max_length  : 901
  beam_size   : 20
  wrote human CSV to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testing/outputs/gravity5unified12345_seed_beam20_beam_human.csv

[beam] n=1
  top-1 correct : 0 / 1 (0.0%)
  any-beam correct: 0 / 1 (0.0%)
  avg valid beams : 20.0

Example predictions (1 rows):
  row 0 [beam]
    target simple : ((e_4 · e_5)*(e_4 · e_5))/6 + ((e_4 · p_5)*(e_4 · p_5)*(e_5 · p_4)*(e_5 · p_4))/(6*(p_4 · p_5)*(p_4 · p_5)) - ((e_4 · p_2)*(e_4 · p_2)*(e_5 · p_3)*(e_5 · p_3))/((p_2 · p_4)*(p_3 · p_5))
    input scram   : ((e_4 · e_5)*(e_4 · e_5))/6 + ((e_4 · p_5)*(e_4 · p_5)*(e_5 · p_4)*(e_5 · p_4))/(6*(p_4 · p_5)*(p_4 · p_5)) - ((e_4 · p_2)*(e_4 · p_2)*(e_5 · p_3)*(e_5 · p_3))/((p_2 · p_4)*(p_3 · p_5))
    top1 pred     : (p_2·F_4·p_5*p_1·F_4·p_5*p_1·F_5·p_4*p_1·F_5·p_4)/(p_1·p_4*p_1·p_5*p_2·p_3*p_4·p_5*p_4·p_5*p_4·p_5)
    top1 metrics  : token=0 string=0 num_simple=0 num_scr=0
    any-beam      : token=0 string=0 num_simple=0 num_scr=0
  wrote detail results to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testing/outputs/gravity5unified12345_seed_beam20_beam_results.csv

Wrote summary to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testing/outputs/gravity5unified12345_seed_beam20_summary.csv
Total wall time: 2.54s

Terminal output PDF: /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testing/outputs/Gravity_5_Evaluation_Information_2026-09-22_23-03-30-382315.pdf

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